

Large deformation measurement scheme for 3D digital image correlation method

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ABSTRACT

Difficulties often arise for digital image correlation (DIC) technique when serious de-correlation occurs between the reference image and the deformed image due to large deformation. An updating reference image scheme could be employed to deal with large deformation situation, however that will introduce accumulated errors. A large deformation measurement scheme, combining improved coarse search method and updating reference image scheme, is proposed in this paper. For a series of deformation images, the correlation calculation begins with a seed point and spreads out. An improved coarse search method is developed to calculate the initial correlation parameters for the seed point, which guarantees that the correlation calculation can be carried out successfully even in large deformation situation. Only for extremely large deformation, the reference image is updated. Using this method, not only extremely large deformation can be measured successfully but also the accumulated error could be controlled. A polymer material tensile test and a foam compression test are used to verify the proposed scheme. Experimental results show that up to 450% tensile deformation and 83% compression deformation can be measured successfully.

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1. Introduction

Digital image correlation (DIC) method [1] can provide full-field displacement and surface strain field of measurement object, by comparing two object digital images before and after deformation. 2D DIC [2], which is used with a single camera, can measure only in-plane displacement/strain fields on plane objects, while 3D DIC [3] technique, which combines DIC with stereo vision, can measure the 3D displacement field and surface strain field of 3D object. Due to its advantages of simple equipment, high precision and non-contact measurement, the DIC method has been used widely for deformation measurement these years. Sztefek et al. [4] used the DIC method to determine the surface strain of bone during loading. Stanford et al. [5] made use of the DIC method to measure the wing deformation of micro air vehicles and the measurement result is compared with the numerical result. Miehe et al. [6] applied the DIC method to the surface strain measurement of amorphous glassy polymers. The DIC method is employed in Bambach's research [7] to capture the full-surface transverse buckling deformation of the flanges and webs and the results are used to describe and explain the fundamental behavior of flanges and webs. Experiments based on DIC method were

performed by Hargather and Settles [8] to measure the deformation of thin plates in response to explosive blast.

In recent years, much study has been carried out on DIC method for improving its accuracy, robustness and computational efficiency. Various sub-pixel registration algorithms [9] are developed, and the performance of different methods is discussed. The effect of subset size used in DIC has been investigated, and a theoretical model has been derived for selecting proper subset size to improve measurement accuracy [10]. The impact of interpolation phase error is studied by Schreier et al. [11] and a high-order interpolation scheme is highly recommend by them. Lu and Cary [12] proposed a second-order mapping function, which makes the measurement more accurate when higher order deformation occurs. Pan [13] developed a reliability-guided DIC (RG-DIC) that makes the correlation calculation more robust. A two-step extended DIC (X-DIC) was developed by Chen et al. [14] to measure the full-field displacements across the crack domain. A combined DIC approach and an adaptive DIC approach suitable for samples with discontinuities/holes were proposed by Zhang et al. [15]. More recently, Pan and Li [16] proposed a fast DIC method, whose speed is about 120–200 times faster than that of the traditional method, without any loss of measurement accuracy.

The DIC method uses the random speckle pattern to match the corresponding points in the reference image and the deformed image. In order to track the location of a point in the deformed

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image, a square subset surrounding the point in the reference image is chosen and used as the reference subset. A certain correlation coefficient is predefined to evaluate the similarity between the reference subset and the target subset. Then the corresponding point can be obtained by searching the maximum (or minimum) correlation coefficient in the deformed image. Difficulties often occur when the serious de-correlation happens between the reference image and deformed image because of large deformation. Direct use of DIC method can only deal with deformation up to about 20%, because larger deformation makes the initial estimation of correlation parameters become difficult and leads to the failure of correlation calculation. A technique is presented by Pan et al. [17] that can provide reliable initial value for the first calculation point by picking three corresponding points manually. The method needs human interaction, hence is inefficient if the number of deformation images is large.

In the deformation measurement process, usually a series of deformation images will be captured. Generally the deformation of neighboring stages is continuous. Obviously, an updating reference image scheme [18,19] could be used to deal with large deformation situation. However this method will inevitably introduce accumulated errors. To solve this problem, a large deformation measurement scheme for 3D DIC is put forward in this paper. After selecting calculation area and dividing it into subsets in the reference image, an arbitrary subset is used as seed point. The correlation calculation begins with the seed point, and spreads out until all the points are matched. An improved coarse search scheme, which makes use of the deformation continuity of the adjacent stages, is proposed to provide the initial correlation parameters for the seed point. Using this method, the correlation calculation can be conducted successfully even in large deformation situation (e.g. up to 300% tensile deformation). Only for extremely large deformation (e.g. > 300% tensile deformation), the reference image is re-chosen. By combining the improved coarse search method and updating reference image scheme, not only extremely large deformation can be measured, but also the accumulated error is minimized. A polymer material tensile test and a foam compression test are used to verify the proposed scheme. The results clearly demonstrate that the present technique is reliable and can be applied to extremely large deformation measurement.

2. Key technologies for 3D DIC

2.1. Stereovision for 3D DIC

Stereo vision [20] is a technique for building a 3D description of scene from two different viewpoints. As shown in Fig. 1, $O_w X_w Y_w Z_w$ is the world coordinate system, $O_1 X_1 Y_1 Z_1$ is the left camera coordinate system, $O_2 X_2 Y_2 Z_2$ is the right camera coordinate system, and OXY is the image coordinate system. $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are two corresponding image points of object points $P(x_w, y_w, z_w)$ in the two cameras. The 3D coordinates of object point P can be obtained by triangulation [21] method if (a) the stereo cameras are calibrated and (b) image coordinates of P_1 and P_2 are known. The XJTUDIC system [22] (See Fig. 2) developed by Xi'an Jiaotong University is used in this research.

2.2. Stereo camera calibration

Camera calibration is an important task for 3D DIC method because it directly determines the 3D reconstruction accuracy. Stereo camera calibration involves determining two cameras' interior orientation parameters (the coordinates of principle point x_0, y_0 , the principle distance f and the lens distortion parameters)

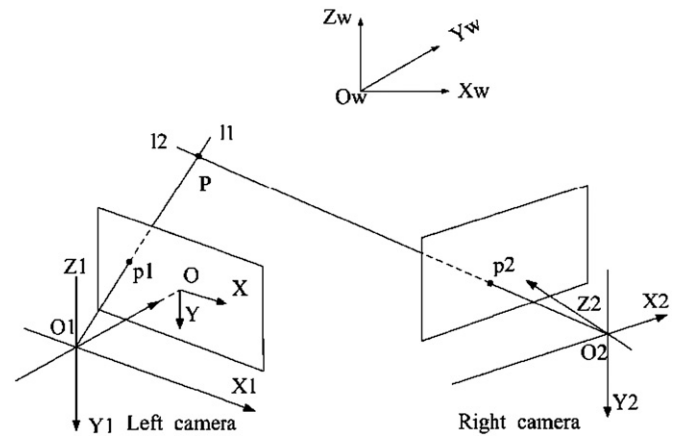


Fig. 1. Stereo vision model.

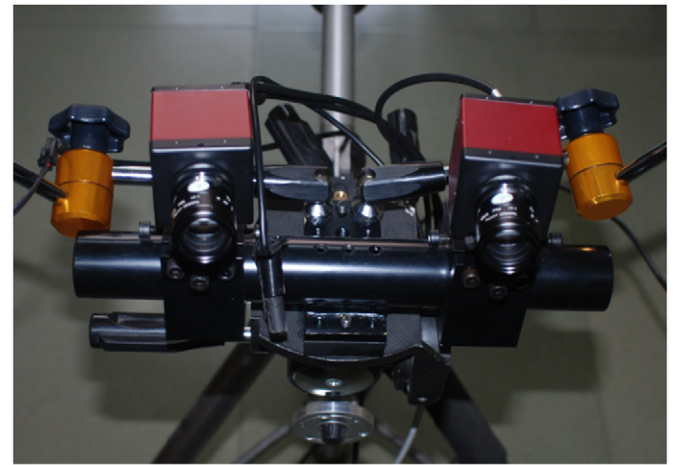


Fig. 2. Construction of stereo vision.

and the relative orientation between two cameras (rotation matrix R and translation matrix T). A flexible and high accuracy self-calibration technique [23] based on photogrammetry has been developed to calibrate the stereovision system. In the calibration process, the two cameras only need to capture a planar target (see Fig. 3) from several different views (usually eight positions). Using the bundle adjustment technique, a reliable calibration result can be achieved without knowing the accurate dimension of the calibration target. Besides, in order to improve the calibration precision, a more complete 10-parameter lens distortion model [24] is applied in the calibration algorithm.

2.3. Basic principle of digital image correlation method

The DIC method uses the random speckle pattern to match the corresponding points precisely on two images. As shown in Fig. 4, in the reference image, a square reference subset of $(2M+1) \times (2M+1)$ pixel centered at point C is picked. The matching procedure is to find the corresponding subset centered at point C' in the deformed image, which has the maximum similarity with the reference subset. Consequently, the two center points C and C' are a couple of corresponding points of the two images. A predefined cross correlation (CC) criterion or sum of squared differences (SSD) criterion [25] is used to evaluate the similarity between the reference subset and the target subset. So the corresponding subset location can be determined by searching

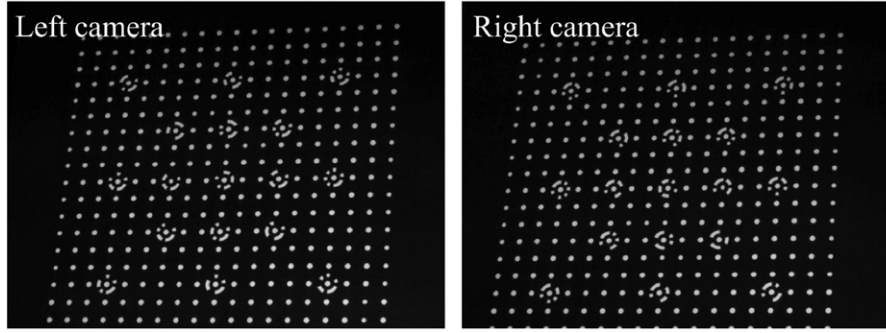


Fig. 3. A pair of calibration target images used in the calibration.

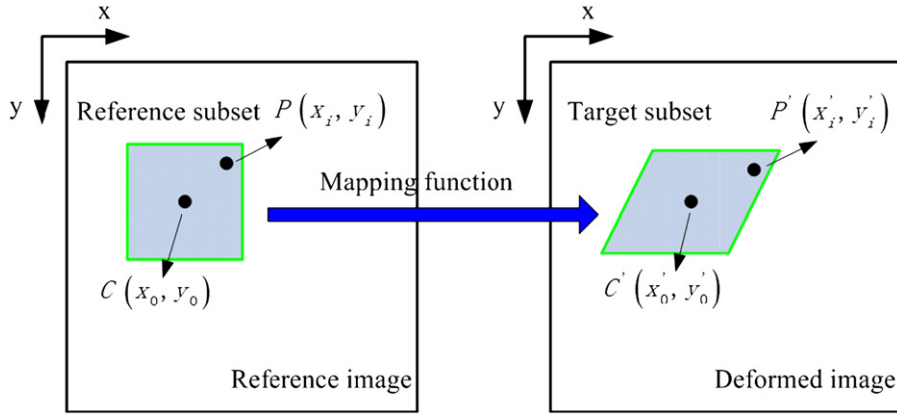


Fig. 4. Basic principle of digital image correlation.

the maximum or minimum (according to the used correlation criterion) correlation criterion in the specified searching area.

Due to its simplicity, the SSD correlation is adopted in this work:

$$C_{SSD}(p) = \sum_{x=-M}^{x=M} \sum_{y=-M}^{y=M} [f(x,y) - r_0 - r_1 * g(x',y')]^2 \quad (1)$$

where $f(x,y)$ is the gray value of point (x,y) in the reference subset of reference image, $g(x',y')$ is the gray value of point (x',y') in the corresponding subset of deformed image, r_0, r_1 are used to compensate the gray value difference caused by illumination diversity, $p=[u, u_x, u_y, v, v_x, v_y, r_0, r_1]$ represents the vector of the correlation parameters that depends on the chosen mapping function. The first order mapping function is used in this work:

$$\begin{aligned} x'_i &= x_0 + \Delta x + u + u_x \Delta x + u_y \Delta y \\ y'_i &= y_0 + \Delta y + v + v_x \Delta x + v_y \Delta y \end{aligned} \quad (2)$$

where $\Delta x, \Delta y$ are the distances from the subset center to point (x_i, y_i) , u and v are the displacement components of the reference subset center in x and y directions, and u_x, u_y, v_x, v_y are the first-order displacement gradients of the reference subset.

To get the minimum of C_{SSD} is a nonlinear minimization problem, which can be solved by using the ILS algorithm [26]. Still it needs to be noticed that a bi-cubic spline interpolation [27] is adopted in the algorithm realization because the coordinates of points in the deformed image are not integer pixel. Moreover, to evaluate the accuracy of ILS results, the standard deviation can be employed:

$$s_0 = \sqrt{\frac{\sum \sum v^2}{n-1}} \quad (3)$$

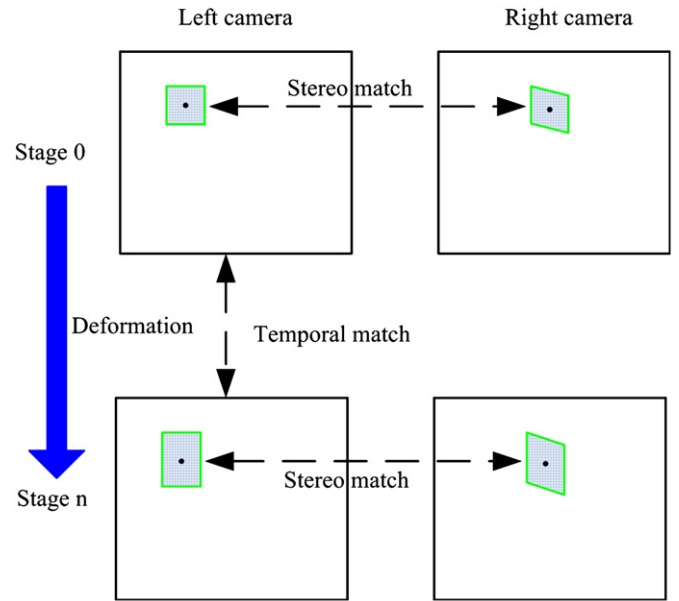


Fig. 5. Match procedure of 3D DIC.

where n is the number of pixels in the reference subset, v represents the gray value residual of one pixel after correlation calculation.

2.4. The overall procedure of 3D DIC

Fig. 5 displays the match procedure of 3D DIC. The match of 3D DIC consists of stereo match and temporal match. For stereo match,

the epipolar constraint could be used to improve the match efficiency [28]. However, when using the epipolar constraint, the DIC algorithm may be constrained to find an erroneous match instead of the best match [29]. This is because the calibration error of stereo system will inevitably be introduced to the match result. To avoid this problem, in our method, the epipolar constraint is only used to find the initial match result during coarse search in pixel resolution, yet not used when calculating the final match result in sub-pixel resolution using the ILS algorithm. For temporal match, difficulty often occurs in large deformation situation. After all the images are matched, for each stage, 3D coordinates of all the point can be obtained through triangulation method using the calibration parameters of the stereo cameras and the corresponding image points in the left and right images.

If the 3D reconstruction of all the stages is finished, the 3D displacement of any point in a stage can be obtained directly by comparing its 3D coordinates in current stage and the reference stage.

Furthermore, using the nine neighbor points' coordinates, the deformation gradient tensor F for any point can be calculated and accordingly the strain can be obtained.

3. Proposed large deformation measurement scheme

3.1. Image correlation method based on seed point

The ILS algorithm is a non-linear optimization method, and therefore how to find reliable initial value for ILS algorithm is a key problem, especially in large deformation situation. In order to get the initial value for the ILS algorithm, a pixel by pixel coarse search method [30] is usually used to find the initial displacement component (u , v). Because the coarse search method is quite time-consuming and only displacement component is considered that may decrease the convergence speed of ILS algorithm, the whole match procedure will be quite inefficient if the initial value of each point is provided through this method.

Usually an image correlation method based on seed point could be utilized to speed up the correlation calculation speed. An arbitrary subset is chosen as the seed point and matched first. After the seed point is matched successfully, the correlation parameters of the seed point could be used as the initial guess of its four neighbor points according to deformation continuity assumption. The process repeats until all the points are matched successfully. Because the coarse search method is only used for the seed point, the whole match process time is much reduced.

Still there is another issue that the coarse search method often fails for temporal match when large deformation happens. To overcome this problem, an improved coarse scheme, which

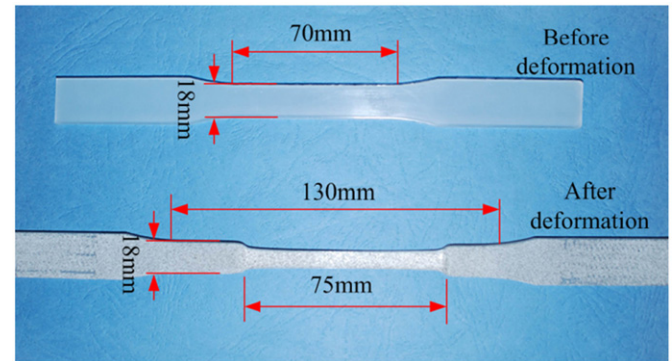


Fig. 7. Polymer specimen before and after deformation.

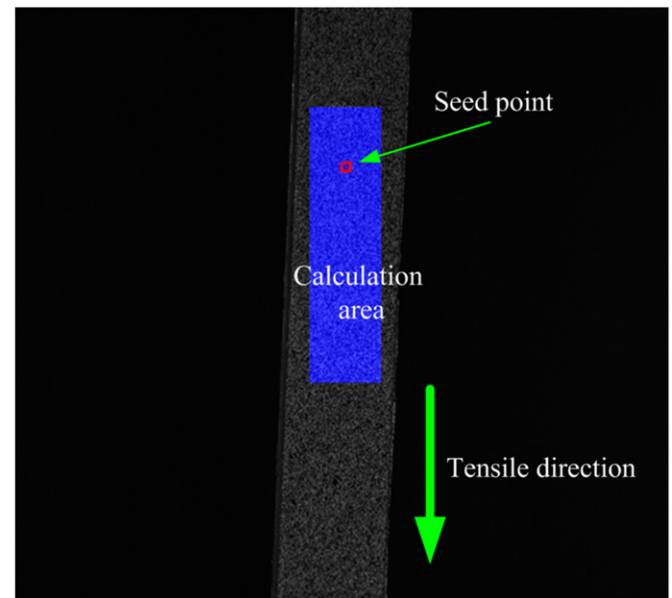


Fig. 8. Calculation area and seed point in the reference image.

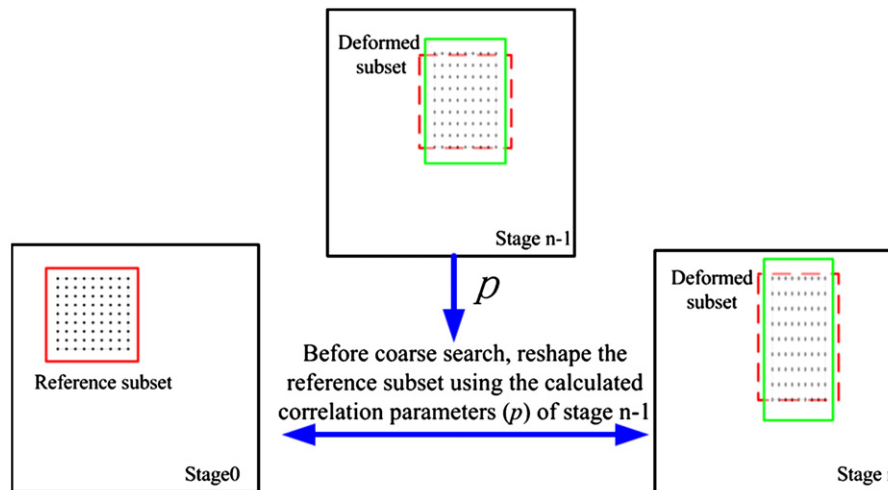


Fig. 6. Schematic diagram of improved coarse search method.

makes use of the deformation continuity of the adjacent states, is described in the following section.

3.2. Improved coarse search method

In traditional coarse search method, the shape of the reference subset is assumed to remain unchanged in the deformed image and only translation occurs. In order to obtain the initial displacement value, the reference subset is moved pixel by pixel in the specified search area of the deformed image and the correlation coefficient is calculated. The location that has the maximum or minimum correlation coefficient is the target subset. The coarse search method works well when the relative deformation is small ($< 20\%$). However in large deformation situation, the similarity between the reference subset and deformed subset decreases rapidly and the correlation coefficient distribution map does not have a significant peak, which leads to the failure of coarse search method.

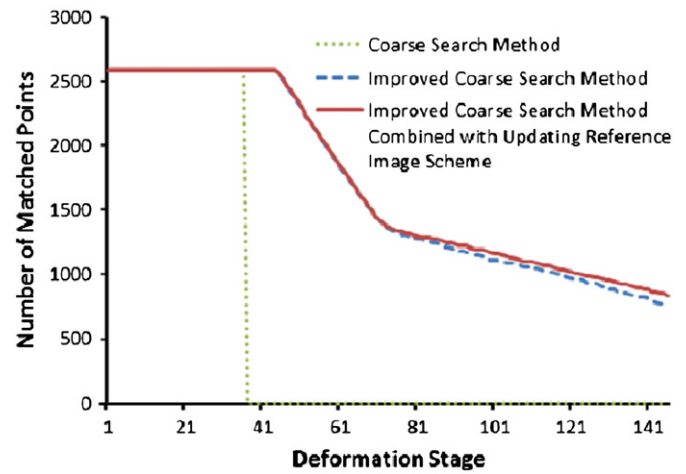


Fig. 11. Number of matched points of all stages using different methods.

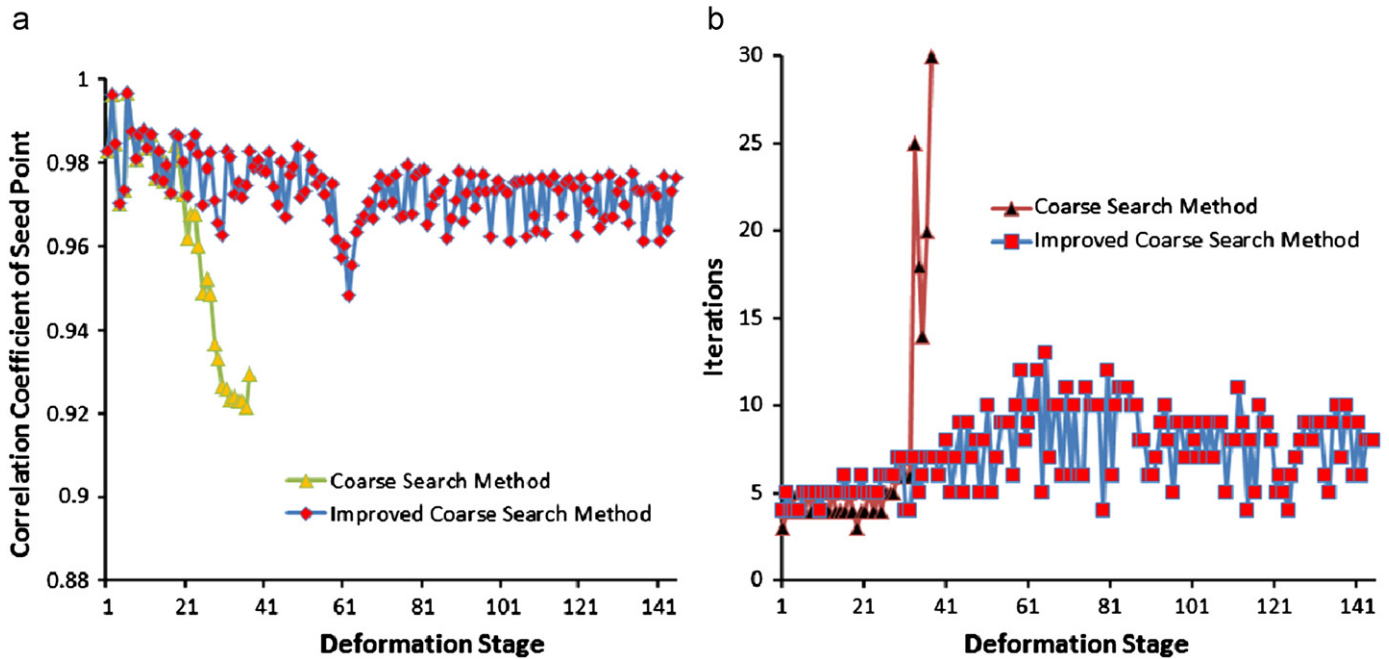


Fig. 9. Calculated correlation coefficient (a) and iterations of ILS algorithm (b) for the seed point using coarse search method and improved coarse search method.

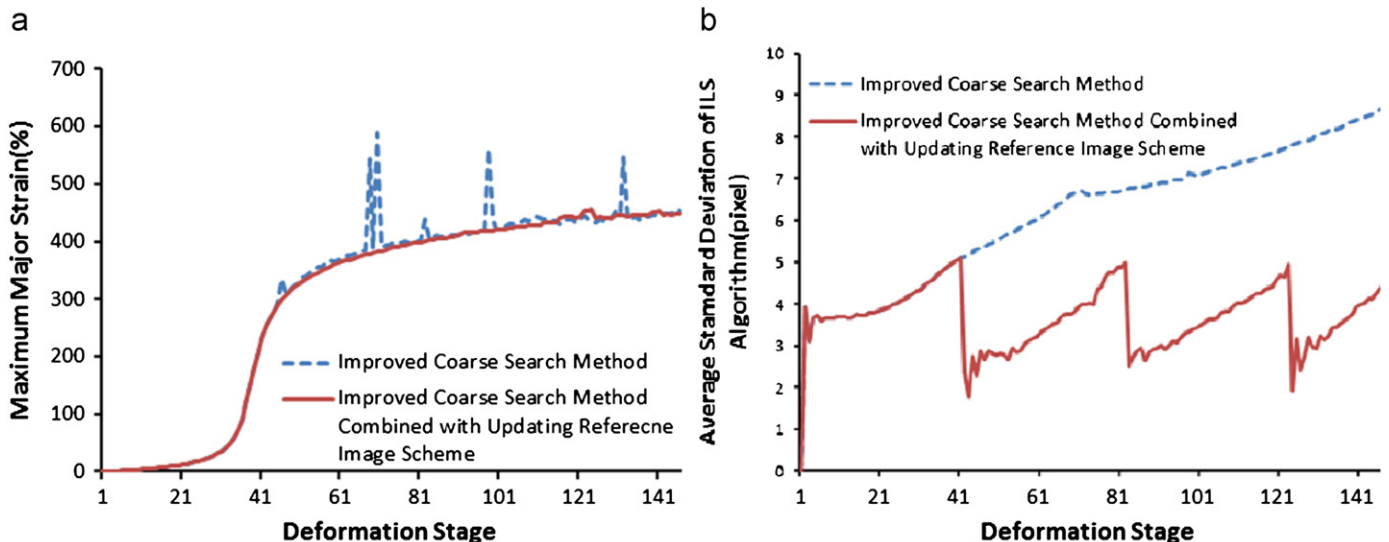


Fig. 10. Maximum major strain (a) and residual of ILS algorithm (b) of different stage using improved coarse search method with and without the updating reference image scheme.

Typically, a series of deformation image (whose deformation becomes larger continuously) will be captured in the deformation measurement process. Using the first image as the reference image, the image of stage n usually has lower similarity with the reference image, but has higher similarity with the image of stage $n - 1$. Of course, updating a reference image scheme could be used to handle the large deformation situation, however will inevitably introduce accumulated errors. So the image of stage

$n - 1$ should be used to assist the correlation calculation between the image of stage n and the reference image but should not be used as reference image directly. Considering this, an improved coarse search method is formed. Unlike the traditional coarse search method, the reference subset is not assumed to maintain its shape in the searching process. As illustrated in Fig. 6, in large deformation situation of stage n , if the same point is matched successfully in the stage $n - 1$, the reference subset is first

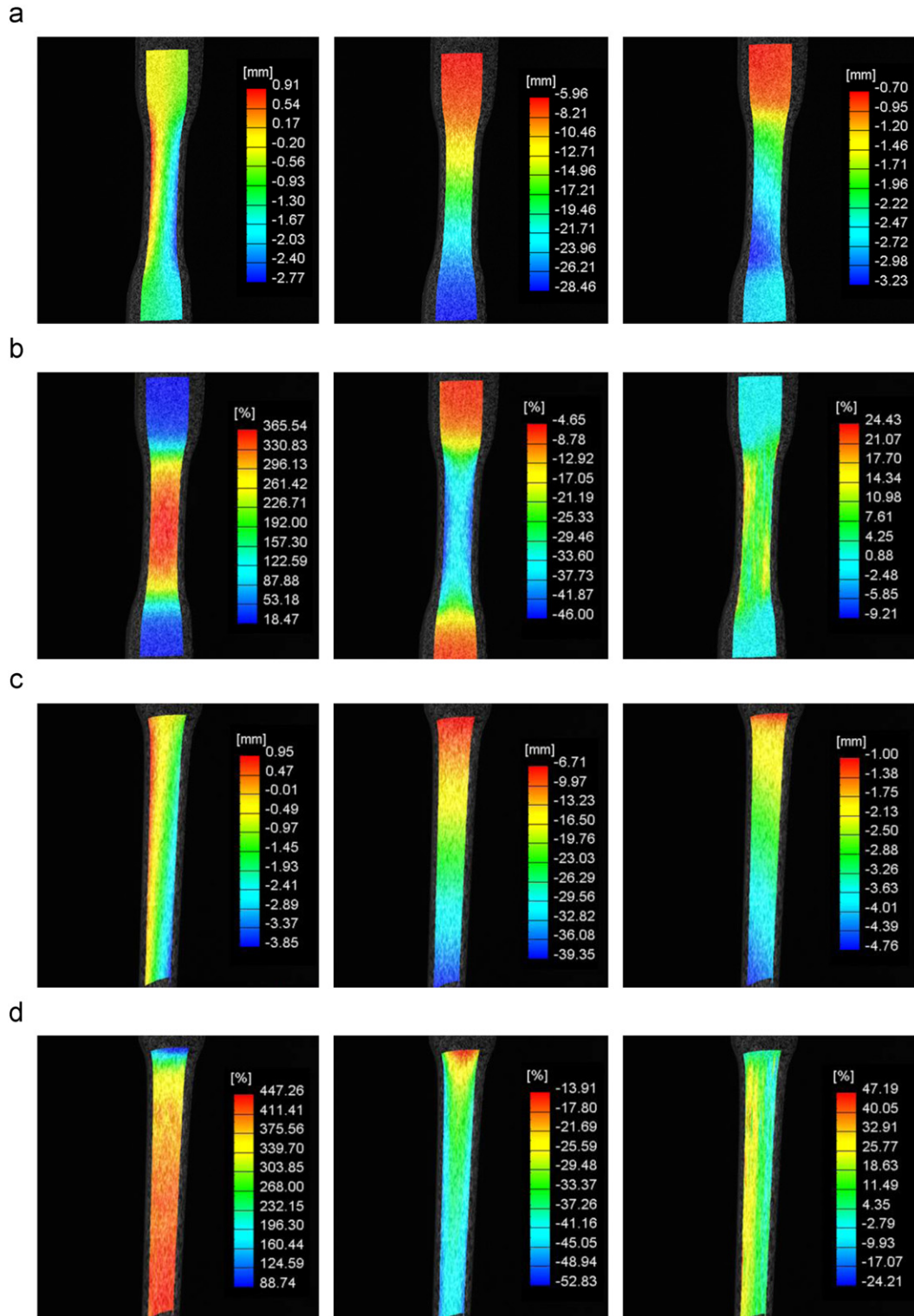


Fig. 12. Displacement fields in x , y , z directions (a, c) and ϵ_x , ϵ_y , ϵ_{xy} strain fields (b, d) of the 60th and 146th stage superimposed on the specimen images.

reshaped using the correlation parameter (p) of stage $n-1$ according to the mapping function (see Eq. (2)). Afterwards the reshaped reference subset is moved in the search area and the correlation coefficient is calculated. Using this method, the correlation coefficient distribution map will have a significant peak, and the subset is matched successfully.

The proposed method could deal with large deformation well (e.g. up to 300% tensile deformation). However, when extremely large deformation occurs, the proposed method may become unstable (e.g. $> 300\%$ tensile deformation). Therefore for extremely large deformation situation, the reference image updating method is applied. The standard deviation of ILS algorithm (Eq. 3) is used to judge whether the reference image needs to be updated, and if average standard deviation of a stage is big than a predefined threshold (e.g. 5 pixel), then the current stage is employed as new reference image. By combining the improved coarse method with the updating reference image method, not only extremely large deformation could be measured successfully, but also the reference image updating times is reduced to the least, hence the accumulated error is controlled.

When using DIC method, the error of image correlation is affected by many factors, including experimental conditions (such as speckle pattern, image distortion, noise and so on) and correlation algorithm (such as subset size, correlation criterion, interpolation, shape function and so on) [2]. Generally, the best image correlation accuracy for DIC method is about 0.02 pixels or better [27]. The above part is about the correlation error of DIC method itself, while the purpose of our work is trying to control the accumulated error introduced by updating reference image scheme. For example, for a series of deformation images, if the error for each DIC calculation is 0.02 pixels and the reference image is updated m times using our method, then in the worst case, the correlation error between stage 0 and stage n is $0.02 \times (m+1)$ pixels.

4. Experiments and discussion

To verify the proposed large deformation measurement scheme, a polymer material tensile test and a foam compression test are carried out.

4.1. Polymer material tensile test

Fig. 7 shows the polymer specimen before and after deformation used in the tensile test. Dimensions of gage region of the specimen is $70 \times 18 \times 4$ mm. Comparing the dimensions of the

specimen before and after deformation, indicates that extremely large tensile deformation ($> 400\%$) occurs in the middle region of the specimen. The tensile test is carried out on the RGM4100 universal test made by RIGER Corporation, and the XJTUDIC system (see Fig. 2) is used to capture the deformation images. 146 stages are collected throughout the experiment.

In the reference image (see Fig. 8), a rectangular area on the specimen is selected as the calculation area and divide it into subsets. The subsets size is set to 15×15 pixels and the subset step is set to 5 pixels. Accordingly 2581 points are calculated in the calculation area altogether. Then a subset is chosen as the seed point and the calculation begins with the seed point. Three different means are employed to do the correlation work, respectively: traditional coarse search method, improved coarse search method with and without updating reference image scheme.

For the seed point, calculation result using traditional coarse search method and improved coarse search method can be seen in Fig. 9. For traditional coarse search method, after the 20th stage, the correlation coefficient (CC criterion is used) decreases seriously and the iteration of the ILS algorithm increases rapidly. The traditional coarse search method can only calculate the first 35 stages, while the improved coarse search method is more stable and could handle all the 146 stages. As a matter of fact, the traditional coarse method relies heavily on the choice of seed point, and a better result can be obtained if the chosen seed point undergoes small deformation. In contrast, the improved coarse search method is seldom affected by the seed point choice and large deformation seed point could also be well handled.

After the seed point is matched successfully, the correlation calculation spreads out according to deformation continuity assumption until all the points are matched. After the correlation work is finished, the 3D point coordinates, displacements and strains can then be gained according to the procedure described in Section 2.4. Fig. 10 (a) shows the maximum major strain curve of all the stages obtained using improved coarse search method with and without updating reference image scheme. As we can see from Fig. 10 (a), the improved coarse search method could handle up to 300% deformation successfully, however becomes unstable for extremely large deformation ($> 300\%$). In contrast, combining the improved coarse search method with updating reference image scheme makes the measurement result of extremely large deformation much more reliable. According to the method described in Section 3.2, the reference images only need to be updated three times. In addition, Fig. 10 (b) shows the average standard deviation curve of different stages using improved coarse search method with and without updating

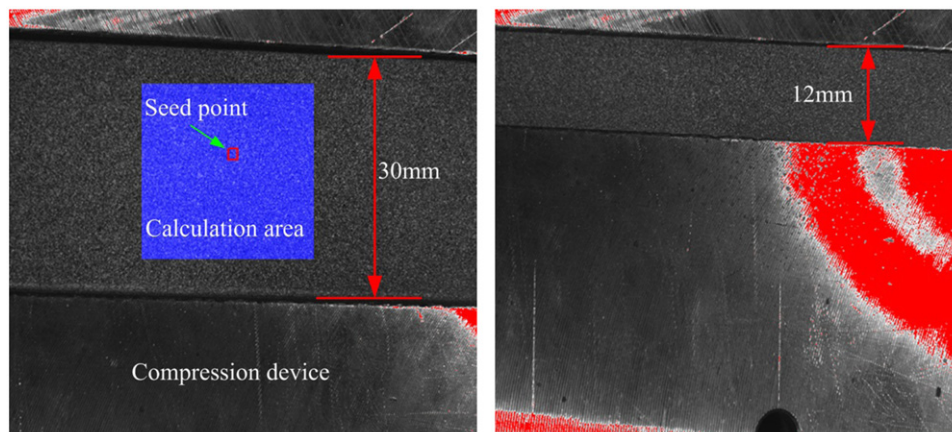


Fig. 13. Foam specimen before (a) and after (b) deformation.

reference image scheme. The updating reference image scheme keeps the standard deviation of the ILS algorithm below 5 pixels, which makes the measurement more stable.

Fig. 11 displays the number of successfully matched points using three different methods. The traditional method can only handle the first 35 stages. Whereas the improved coarse search method could process all the 146 stages, although in extremely large deformation situation the de-correlation effect causes the failure of correlation calculation for few points. Furthermore, to combine the improved

coarse search method with updating reference image scheme makes the calculation result more complete and reliable.

The measurement results including displacement fields (in x , y , z directions) and strain fields (ϵ_x , ϵ_y , ϵ_{xy} strain) of the 60th and 146th stage are illustrated in Fig. 12. For the 146th stage, the maximum displacement in y direction (tensile direction) is about 39 mm, the maximum displacement in z direction (out-of-plane displacement) is about 4.8 mm, and the maximum ϵ_x strain (tensile direction) is up to 447%.

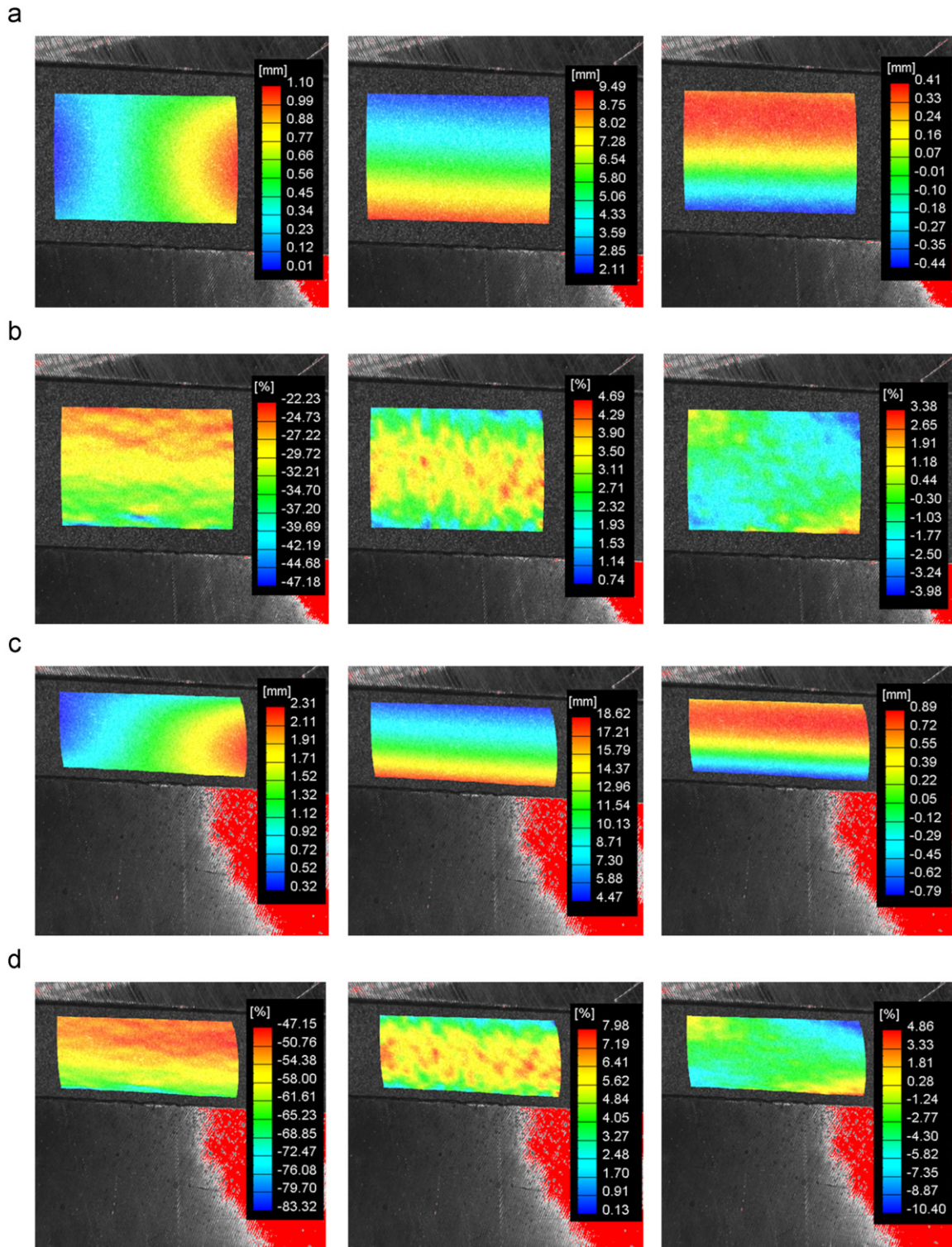


Fig. 14. Displacement fields in x , y , z directions (a, c) and ϵ_x , ϵ_y , ϵ_{xy} strain fields (b, d) of the 60th and 105th stage superimposed on the specimen images.

4.2. Foam compression test

Fig. 13 demonstrates the foam specimen and the compression device before and after deformation. An average 60% compression is imposed on the specimen, thus the specimen height decreases from 30 mm to 12 mm. Totally 105 stages are recorded during the test. The de-correlation effect is found to be more serious for the foam material because of its porous structure. After specifying the calculation area and the seed point (see Fig. 13a), 1860 points with subset of 25×25 pixel and subset step of 13 pixels are calculated. The same process and analysis method employed in the tensile test is performed on the compression test, and similar conclusion is obtained. The improved coarse search method performs much better than the traditional coarse search method, yet combining the improved coarse search method with the updating reference image scheme gives the best performance. According to method described in Section 3.2, the reference image only needs to be updated twice. The measurement results including displacement fields (in x , y , z directions) and strain fields (ε_x , ε_y , ε_{xy} strain) of the 60th and 105th stage are illustrated in Fig. 14. For the 105th stage, the maximum displacement in y direction (compression direction) is about 18.6 mm, the maximum displacement in z direction (out-of-plane displacement) is about 0.9 mm, and the maximum ε_x strain (compression direction) is up to 83%.

5. Conclusion

To solve the problem of large deformation measurement using 3D DIC method, a large deformation measurement scheme, combining improved coarse search method with updating reference image scheme is presented. To improve the calculation speed, a seed point based method is adopted. According to the deformation continuity of adjacent stage, an improved coarse search method is proposed, which guarantees that the correlation calculation can be carried out successfully in large deformation situation (up to 300% tensile deformation). Only for extremely large deformation (e.g. > 300% tensile deformation), the updating reference image scheme is employed. By combining improved coarse search method with updating reference image scheme, not only large deformation can be measured automatically, but also the accumulated error introduced by updating reference image can be controlled. A polymer material tensile test and a foam compression test are carried out to verify the proposed method. Measurement results of up to 450% tensile deformation and 83% compression deformation confirm the validity of the method. The proposed scheme in this work is very useful for the material properties analysis of polymer or foam material that may experience extremely large deformation.

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